

**Continue: Differentiation.**

$\overline{4}$ .

3. If  $y = (\sinh^{-1} x)^2$ , prove that  $(1 + x^2)y'' + xy' = 2$ .
4. If  $y = (\sin^{-1} x)^2$ , prove that  $(1 - x^2)y''' - 3xy'' - y' = 0$ .
5. Find the derivative  $y' = \frac{dy}{dx}$  of the following functions:
  - a)  $y = \cos^3 x - 12 \sin x^2$
  - b)  $y = \tan(8x) + 2 \sec^3(12x)$
  - c)  $y = 4^{(7x^2)} \tan^2(6x)$
  - d)  $y = \cot^3(7x) + \csc^4(6x)$
  - e)  $y = \cos(2x^2) - \cot^3(9x)$
  - f)  $y = (x - 4) \cos^2(9x)$
  - g)  $y = \frac{1 - \tan x}{1 + \tan x}$
  - h)  $y = (x + \cot x)^{\frac{5}{2}}$
  - i)  $y = (\tan x^2)^5$
  - j)  $y = \sqrt{\sin(\sqrt{x})}$
  - k)  $y = \cot^2 x + \cot x^2$
  - l)  $y = \sin(\tan(\sqrt{\sin x}))$
  - m)  $y = \cos(\sqrt{1 + x^2})$
  - n)  $y = \cos^{-1}(\sin^{-1} x)$
  - o)  $y = \sin^{-1}(3x + 1)$
  - p)  $y = \ln x \cdot \sec^{-1}(3x^2)$
  - q)  $y = \sin^{-1}(\tan^{-1} x)$
  - r)  $y = \cot^{-1}\left(\frac{1+x}{1-x}\right)$

$$\text{s)} \quad y = \frac{x + \sec^{-1} x}{x^2 + \tan^{-1} x}$$

$$\text{t)} \quad y = \frac{1}{x^2} + \sin^{-1}(2x - 1)$$

$$\text{u)} \quad y = \tan^{-1}(x^3)$$

$$\text{v)} \quad y = (\cot^{-1} x)^3$$

$$\text{w)} \quad y = \sec^{-1} \sqrt{1 + x^2}$$

$$\text{x)} \quad y = \tan^{-1}\left(\frac{x}{a}\right)$$

$$\text{y)} \quad y = \sqrt{\csc^{-1} x}$$

$$\text{z)} \quad y = (\tan^{-1} x)^{-1} + \tan^{-1}(\sin x)$$

$$\text{i)} \quad y = (x^2 + 1) + \sin^{-1}\left(\frac{x}{a}\right)$$

$$\text{ii)} \quad y = \sqrt{x^2 - 1} + \sin^{-1} x$$

$$\text{iii)} \quad y = \cos(\sqrt{1 + \sin^2 x}) - \csc^{-1}(3x)$$

$$\text{iv)} \quad y = (\sec^2(3x) + \cos^{-1} x)^4$$

$$\text{v)} \quad y = \sin^{-1}\left(\frac{x-1}{x+1}\right)$$

$$\text{vi)} \quad y = \tan^{-1}(\sin^{-1}(\sqrt{x})).$$

$$\text{vii)} \quad y = \cosh^{-4}(\sqrt{2x + 1})$$

$$\text{viii)} \quad y = (\log_2 x) \cdot [\operatorname{sech}(e^{2x})]$$

$$\text{ix)} \quad y = \tan^2(6x - 4)$$

$$\text{x)} \quad y = x^2 \csc^4(6x)$$

$$\text{xi)} \quad y = \cosh^3 x - 12 \sin^{-1} x$$

$$\text{xii)} \quad y = 2 \operatorname{sech}^{-1}(12x)$$

$$\text{xiii)} \quad y = x^3 \coth^2(6x)$$

$$\text{xiv)} \quad y = (\tanh x)^{-1}$$

$$\text{xv)} \quad y = \tanh^{-1}(2x) + \operatorname{csch}^{-1}(2x)$$

## Some Applications on Differentiation.

6. I- Logarithmic Differentiation: Find  $y'$  if:

a)  $y = \frac{\sin^2 x \tanh^4 x}{(x^2+1)^2}$

b)  $y = \frac{(x-1)^3 \sqrt{x-2}}{\sqrt[4]{x-6}}$

c)  $y = \sqrt{x \cdot \sinh^{-1} x \cdot \sqrt{1-e^x}}$

d)  $y = (\sin x)^x$

e)  $y = x^{\ln x}$

f)  $y = (\ln x)^{\cos x}$

g)  $y = (\sinh x)^{\frac{1}{x}}$

h)  $y = (\operatorname{csch}^{-1} x)^2 + (\coth x)^{(\sqrt{x})}$

7. II- Implicit Differentiation: Find  $y'$  if:

a)  $x^2y + 3xy^2 - x = 3$

b)  $\cos(xy) = y$

c)  $\sin(x^2y^2) = x$

d)  $\frac{xy^3}{1+\sec y} = 1 + y^4.$

e)  $x^2y^2 - y \ln x = 10$

f)  $\tan(\sqrt{x}) - y \cdot \sec(\sqrt{y}) = 1.$